

Longevity-favourable foods

The opposite of the intolerant report: 173 useful foods for longevity evaluated by genetics, and only those with favourable genetic indication (⊕) listed, by category
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Genetic basis: Raw DNA of a genotyping chip (GRCh38, 684 967 SNPs) + local imputation (Beagle 5.5, public reference panel of genomes complete to 30x; only genotypes with DR2 ≥ 0.8). Genetic data did not leave the local computer. Only genetics: no analyses or exams were used.

Mechanisms evaluated: 21 Nutrigenomic mechanisms favourable (crucifers × GSTM1/GSTA1/NRF2/NQO1, vegetables and fruit × 9p21, Mediterranean diet pattern × TCF7L2, olive oil × PON1, omega-3 × FADS, fiber × LDL/glucose, folate and riboflavin × MTHFR, betaine × BHMT, choline × PEMT, potassium × salt sensitivity, nitrate × NOS3, zinc × SLC30A8, lutein × CFH, antioxidants × SOD2, Fermented dairy × LCT/HFE, tea × HFE, decaffeinated coffee × liver/CYP1A2) and 1 support note (protein × predisposition to sarcopenia), with about 90 variants verified in the chip and imputation.

List of foods: only useful foods for longevity (crucifers, other vegetables and mushrooms, legumes and soybeans, whole grains, dried fruits and seeds, small fatty fish, berries and other fruit, olive oil, fermented coffee, decaffeinated tea, cocoa, herbs and spices), each linked to the aging mechanisms that touch the most. The same method, panel and format of the intolerance report.

Important: Most of the nutritional evidence is weak to moderate (association studies and secondary trials). "⊕" means that genetics points to a benefit equal or greater than usual, not that food is mandatory or that others are bad. All indications are suggestion (not applied).

Moderate ⊕ at least one mechanism of moderate or weak evidence and several mechanisms to add **Weak ⊕** genetic indication with weak evidence or a single mechanism. On request, the report shows only ⊕ foods; neutral foods are counted but not listed.

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1. Executive summary

Class	Food No.	% of the evaluated
Moderate  favourable genetic indication, moderate evidence / various mechanisms	73	42.2%
Weak  favourable genetic indication, weak evidence	68	39.3%
Neutral (useful for longevity, but without specific genetic indication)	32	18.5%
Total longevity foods evaluated	173	100%
Withdrawal before evaluation (rule or conflict; do not count on the evaluated)	13	—

141 of the 173 evaluated longevity foods have favourable genetic indication; 32 are neutral (not listed).

Fast reading — 18 foods with more genetic mechanisms in favor (key genes between parentheses)

Beak grain — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), FUT2, AGT/ADD1/GNB3 (salt), SLC30A8, PEMT

Lentils — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), FUT2, AGT/ADD1/GNB3 (salt), SLC30A8, PEMT

Bruxella — GSTM1/GSTA1/NRF2/NQO1, 9p21, MTHFR (folate), PEMT, AGT/ADD1/GNB3 (salt), CETP/APOB (LDL)

Edamame — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), PEMT, AGT/ADD1/GNB3 (salt), FADS (ALA)

Soybean — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), PEMT, AGT/ADD1/GNB3 (salt), FADS (ALA)

Spinach — 9p21, NOS3, MTHFR (folate), BHMT, CFH, AGT/ADD1/GNB3 (salt), MTHFR (B2)

Fresh peas — 9p21, MTHFR (folate), CFH, CETP/APOB (LDL), TCF7L2/MTNR1B (glucose)

Avocado — 9p21, AGT/ADD1/GNB3 (salt), MTHFR (folate), CFH, CETP/APOB (LDL)

Artichoke — 9p21, FUT2, MTHFR (folate), AGT/ADD1/GNB3 (salt), CETP/APOB (LDL)

Dry peas — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), FUT2, AGT/ADD1/GNB3 (salt)

Beans — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), FUT2, AGT/ADD1/GNB3 (salt)

Azuki beans — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), FUT2, AGT/ADD1/GNB3 (salt)

White bean — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), FUT2, AGT/ADD1/GNB3 (salt)

Embedded beans — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), FUT2, AGT/ADD1/GNB3 (salt)

Black bean — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), FUT2, AGT/ADD1/GNB3 (salt)

Beans — MTHFR (folate), CETP/APOB (LDL), TCF7L2/MTNR1B (glucose), FUT2, AGT/ADD1/GNB3 (salt)

Broccoli — GSTM1/GSTA1/NRF2/NQO1, 9p21, MTHFR (folate), PEMT, CFH

Noz — FADS (ALA), TCF7L2 (Mediterranean diet), CETP/APOB (LDL), MTHFR (folate)

In summary: The genes favor a Mediterranean pattern of vegetable base: legumes and soybean every day, daily crucifers (of raw or little cooked preference), green leaves and beet, oats and barley, walnut and flaxseed, extra virgin olive oil as main fat, small fatty fish and berries. Most of these signals have a small effect; the most important is the set.

Favourable points and limits, separated

IN FAVOUR — genetics that improves the response

- **Crucifers:** Probable null GSTM1 + GSTA1 *B/*B (isothiocyanates stay longer) with NRF2 –617 C/C and NQO1 *1/*1 intact to respond.
- **Vegetables, fruit and berries:** 9p21 homozygous for the risk haplotype (5 SNPs: 4 chip + 1 imputed, r 0.92–0.998), the genotype in which a diet rich in vegetables and fruit attenuated the risk of infarction.
- **Extra virgin olive oil and dry fruits:** TCF7L2 C/T (the Mediterranean diet attenuated its effect) and PON1 of low expression (oil polyphenols rise to PON1).
- **Omega-3** FADS of high activity, which converts well the ALA and makes the preformed EPA/DHA useful; PNPLA3 148M.
- **Legumes, leaves and germ of wheat:** MTHFR compound + high genetic homocysteine (folate, B2, betaine).
- **Collin without cholesterol** (soybean, fish, crucifers, quinoa): PEMT Met175 and genetic liver.
- **Viscous fiber:** High CETP, Polygenic ApoB above average and 9p21: plus LDL/ApoB below.
- **Potassium, nitrate, zinc and lutein:** sensitivity to salt, NOS3 –786C/Asp298, SLC30A8 A/A and CFH 402H heterozygous.
- **Strong tolerances that help:** persistent lactase (yogurt, kefir), without HLA-DQ2/DQ8 (wheat, rye, barley), normal G6PD (fava beans).

LIMITS — what reduces, conditions or does not know

- **Evidence:** almost all gene x food interactions come from association studies or secondary analyzes; none is a formal clinical recommendation.
- **COMT Val/Val** methyl faster catechins and flavanols (green tea, cocoa): less exposure, so cocoa is neutral.
- **Insomnia p89 (CYP1A2 *1F A/C of normal elimination; ADORA2A C/C less sensitive:** coffee with caffeine is out of this list (it is in "limiting"); tea counts for caffeine of the day.
- **Salt sensitivity:** the salted fermented (miso, kimchi, pickles, olives in brine) are left out.
- **HFE H63D:** Iron-rich foods (starch, mussel, viscera, red meat) do not enter; oysters only occasionally.
- **Histamine (DAO/HNMT heterozygotes):** Fat fish, spinach, avocado, tempeh, natto and kefir are in "watch": fresh and well preserved.
- **Genetic liver (PNPLA3, MBOAT7, Pi*MZ):** sweeter fruit in moderate portion; zero alcohol (alcoholic beverages were not evaluated).
- **TCF7L2 x fiber:** In a cohort, fiber protection against diabetes was clearer in non-carriers; fiber remains always better by LDL and microbiome.
- **Untested:** GSTT1, copy number of GSTM1 and AMY1, microbiome.

Conflicts with the report of intolerances (left out)

Some longevity foods are in "limit" in the report of intolerances and, as a rule, do not appear here, even if they have benefits: coffee with caffeine (filtered or expressed) and tea matcha, by caffeine (by sleep, insomnia p89; CYP1A2 *1F A/C of normal elimination and less sensitive ADORA2A C/C; filtered coffee protects the liver, but the dose of caffeine is what orders: up to ~200 mg, only in the morning), and fermented and preserved salt (miso, kimchi, fermented pickles, olives in brine, smoked salmon), by salt. Alternatively, the filtered decaffeinated coffee enters as ☉. Grapefruit and pomelo (interaction with drugs metabolized by CYP3A4), Brazil nuts (excess of selenium), clams (very high iron), cod liver oil (retinol; already taking fish oil) and all alcoholic beverages (zero alcohol) were also excluded. The ☉ foods that are in "watch" in the intolerance report have this note in line.

2. Favourable nutritional mechanisms evaluated

Mechanism	Genes	Genotype (strand +, GRCh38)	Interpretation	Evidence	Trust	Mechanisms of ageing
Isothiocyanates / sulforaphane (crucifers) 20 ⊕ foods	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38	Probable null GSTM1 (rs366631 A/A); GSTA1 *B/*B (rs3957357 A/A); NFE2L2 –617 C/C (rs6721961 G/G, imputation DR2 1.00); NQO1 *1/*1 (rs1800566 G/G, rs1131341 G/G); GSTP1 Ile105Val (rs1695 A/G); TAS2R38 heterozygous (rs713598 C/G, rs1726866 G/A, rs10246939 T/C)	Conjugates the slower isothiocyanates (they are in circulation longer) and has the NRF2 → NQO1 pathway intact to respond to them; sulforaphane induces phase II enzymes that are the genetic weak point. In several case-control studies, the protection of crucifers was higher in null GSTM1 (non-unanimous results). Intermediate taster: the bitter is tolerable.	Moderate-weak	chip + imputation (GSTM1 is CNV, inferred)	Oxidative stress; Redox imbalance; DNA Damage and DNA Repair Deficit; Chronic Inflammation
Vegetables, fruit and berries × 9p21 65 ⊕ foods	CDKN2B-AS1 (9p21)	rs1333049 C/C, rs10757278 G/G, rs4977574 G/G, rs1333042 G/G (chip) + rs1556516 C/C (imputed, DR2 1.00): homozygous haplotype of 5 SNPs (r 0.92–0.998)	In a study with two cohorts, a diet rich in raw vegetables, green leaves, fruit and berries attenuated the risk of infarction associated with 9p21; in risk homozygotes with a prudent diet, the risk was close to that of non-carriers. The locus 9p21 regulates CDKN2A/B (p16/p15), central genes of senescence.	Moderate-weak	5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)	Vascular Aging; Arterial stiffness; Cell Senescence
Mediterranean pattern (extra virgin rye, dry fruits) × TCF7L2 4 ⊕ foods	TCF7L2	rs7903146 C/T + rs12255372 G/T (1 allele risk in each)	In a large Mediterranean-diet trial, in T allele carriers the diet with extra virgin olive oil or dried fruits attenuated the effect of TCF7L2 on glucose, lipids and stroke (secondary analysis; the most clear effect was on TT).	Moderate-weak	chip	Hyperglycemia and Intolerance to Glucose; Insulin resistance; Vascular Aging
Polyphenols of low-expression olive oil × PON1 1 ⊕ foods	PON1	Q192Q (rs662 T/T); M55M (rs854560 T/T, imputed DR2 1.00); –108 T/T (inferred)	Little PON1 in serum (Q192 isoform is the most effective against oxidized LDL). Polyphenol-rich oils increase PON1 activity and protect LDL from oxidation. Previous genetic reading: oleic acid with lower levels.	Weak	chip + imputation	Oxidative stress; Vascular Aging; Inflammaging
Omega-3 long chain preformed (EPA/DHA) 10 ⊕ foods	FADS1/FADS2, ELOVL2, PNPLA3, IL6	FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A (high activity); ELOVL2 rs953413 G/G; PNPLA3 I148M C/G; IL6 –174 G/G	FADS of high activity transforms a lot of linoleic acid into arachidonic (proinflammatory precursor); the preformed EPA/DHA balances the omega-6/omega-3 ratio. Previous genetic reading: DHA with increased need. In patients with PNPLA3 148M, an omega-6/omega-3 high ratio was associated with more fat in the liver.	Moderate-weak	chip	Inflammaging; Chronic Inflammation; Vascular Aging; Neuroinflammation
Omega-3 plant (ALA) 8 ⊕ foods	FADS1/FADS2	FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A	Without the alleles that reduce conversion: converts ALA into EPA more efficiently than those carrying smaller alleles. The clinical benefit of this difference is less clear than the conversion itself.	Moderate-weak	chip	Inflammaging; Vascular Aging
Viscous fiber and food "portefolio" × LDL/ApoB 39 ⊕ foods	CETP, APOB, 9p21, ApoB polygenic	CETP TaqIB G/G (B1B1) + rs3764261 C/C (high activity); APOB rs693 G/A; Risk homozygous 9p21; Polygenic score of ApoB at the 67th percentile	β-glucan, legumes, dry fruits, soybean and viscous fiber lower LDL/ApoB in all people (strong evidence). The genetic profile (high CETP, polygenic ApoB above average, 9p21) increases what is to be gained. The gene × fiber interaction itself is not demonstrated. LDL/ApoB lower is better; more fiber is always better.	Moderate	chip + imputation	Vascular Aging; Arterial stiffness; Dysbiosis; Inflammaging

Mechanism	Genes	Genotype (strand +, GRCh38)	Interpretation	Evidence	Trust	Mechanisms of ageing
Low-glycemic intact hydrates 27 ⊕ foods	TCF7L2, MTNR1B, GCKR	TCF7L2 rs7903146 C/T, rs12255372 G/T; MTNR1B rs10830963 C/G; GCKR rs1260326 T/C	Intermediate glycemic response (TCF7L2) and worse at night (MTNR1B). Legumes and intact cereals, rich in fiber, reduce the glycemic load without taking hydrates of quality.	Moderate-weak	chip	Hyperglycemia and Intolerance to Glucose; Insulin resistance; Accumulation of Advanced Glycation Final Products (AGEs)
Prebiotic fiber × secretory FUT2 25 ⊕ foods	FUT2	W154X rs601338 G/A (secretor)	The secretors express glycans in the mucosa that sustain bifidobacteria; the prebiotic fiber (inulin, GOS, β-glucan, resistant starch) tends to give more bifidobacteria and butyrate. Evidence of limited interaction.	Weak	chip	Dysbiosis; Intestinal Barrier dysfunction; Chronic Inflammation
Natural folate × MTHFR 37 ⊕ foods	MTHFR, MTR, SLC19A1	MTHFR C677T G/A + A1298C T/G (probable compound heterozygous); MTR rs1805087 A/A; SLC19A1 rs1051266 T/C; genetic score of homocysteine at the 92nd percentile	With reduced MTHFR activity, food folate lower than homocysteine (light effect on 677TT, lower in compound heterozygotes). Food folate, not synthetic folic acid.	Moderate-weak	chip	Epigenetic Changes; Epigenetic Derivative; Genomic instability; Vascular Aging
Riboflavin (B2) × MTHFR 677T 12 ⊕ foods	MTHFR	C677T rs1801133 G/A	MTHFR 677T loses the cofactor FAD easily; riboflavin stabilizes it. Benefits demonstrated mainly in 677TT (homocysteine); in heterozygous the effect is small.	Weak	chip	Epigenetic Changes; Vascular Aging
Betaine × BHMT/ GNMT 12 ⊕ foods	BHMT, GNMT	BHMT R239Q rs3733890 G/A; GNMT rs9296404 T/T (Imputed DR2 1.00); previous genetic reading: betaine with lower levels	Betaine gives an alternative route (BHMT) to remethylate homocysteine, useful with compound MTHFR. For food, without the effect of increasing the LDL of high doses of supplement.	Weak	chip + imputation	Epigenetic Changes; Vascular Aging; Metabolic Homeostasis Decline
Choline without cholesterol × PEMT and genetic liver 21 ⊕ foods	PEMT, CHDH, PNPLA3, MBOAT7	PEMT V175M rs7946 C/T; PEMT –744 rs12325817 C/C; CHDH rs12676 C/C; PNPLA3 I148M C/G; MBOAT7 rs641738 C/T	Met175 reduces endogenous synthesis of phosphatidylcholine; choline helps the liver to export fat. With the most vulnerable genetic liver (PNPLA3 + MBOAT7), it is convenient to come from sources without much cholesterol.	Weak	chip	Metabolic Homeostasis decline; Endoplasmic Reticulum Disorder; Dysfunction of adipose tissue
Potassium × salt sensitivity 26 ⊕ foods	AGT, ADD1, GNB3, CYP11B2	AGT M235T rs699 A/G; ADD1 G460W rs4961 G/T; GNB3 825 rs5443 C/T; CYP11B2 –344 rs1799998 A/A	Polygenic profile of salt sensitivity; food potassium counteracts the effect of salt on people sensitive to salt. Previous genetic reading: potassium with increased need.	Weak	chip	Vascular Aging; Arterial stiffness
Food nitrate × NOS3 9 ⊕ foods	NOS3	–786 T/C (rs2070744 C/T); Glu298Asp (rs1799983 G/T, imputed DR2 1.00)	Less endothelial production of nitric oxide; the nitrate of vegetables gives NO via nitrate → nitrite → NO, which does not depend on eNOS. Genotype-specific benefit only.	Weak	chip + imputation	Vascular Aging; Arterial stiffness; Oxidative Phosphorylation Decline
Zinc × SLC30A8 13 ⊕ foods	SLC30A8	rs13266634 C/C (Arg325); rs11558471 A/A	In a meta-analysis of 14 cohorts, more zinc was associated with lower fasting glucose, with greater effect on carriers of the rs11558471 A allele (with 2 copies). Previous genetic reading: zinc with increased need.	Weak	chip	Insulin resistance; Immunosenescence; Thymic Involution

Mechanism	Genes	Genotype (strand +, GRCh38)	Interpretation	Evidence	Trust	Mechanisms of ageing
Lutein/zeaxanthin × CFH Y402H 8 ⊕ foods	CFH, ARMS2	CFH rs1061170 C/T (1 allele 402H); ARMS2 rs10490924 G/G (without risk)	1 allele of macular degeneration in CFH; lutein and zeaxanthin are associated with less advanced macular degeneration. The specific effect of the genotype is uncertain.	Weak	chip	Oxidative stress; Inflammaging
Food antioxidants × SOD2 Ala/Ala + GPX1 Leu 13 ⊕ foods	SOD2, GPX1, SELENOP	SOD2 Ala16Ala (rs4880 G/G); GPX1 Pro198Leu (rs1050450 G/A); SELENOP rs3877899 C/T	SOD2 Ala16 enters more mitochondria and generates more H2O2, than GPX1 Leu removes a little worse; in interaction studies, the Ala/Ala with much fruit and vegetables had less risk than the Ala/Ala with little.	Weak	chip	Oxidative stress; Mitochondrial dysfunction; Redox imbalance
Fermented dairy × persistent lactase and HFE 3 ⊕ foods	LCT/MCM6, HFE	LCT −13910 rs4988235 G/A + −22018 rs182549 C/T (persistence); HFE H63D rs1799945 C/G	Digesate lactose (strong evidence for tolerance). Calcium at meals reduces iron absorption, which is useful with H63D and ferritin target below 100. Benefit of longevity of fermented: moderate; the genetic part is weak.	Weak	chip (2 SNPs in agreement)	Dysbiosis; Osteoporosis; Sarcopenia
Iron absorption inhibitors at meals × HFE H63D 3 ⊕ foods	HFE, TMPRSS6	HFE H63D rs1799945 C/G (without C282Y or S65C); TMPRSS6 rs855791 A/G	Tea tannins at meals greatly reduce the absorption of non-heme iron; with H63D heterozygous and ferritin target below 100, this helps. The specific benefit of the genotype is small.	Weak	chip	Oxidative stress; Redox imbalance
Decaffeinated coffee filtered × genetic liver + sleep 1 ⊕ foods	PNPLA3, MBOAT7, SERPINA1, HSD17B13, CYP1A2, ADORA2A	PNPLA3 I148M C/G; MBOAT7 C/T; Probable SERPINA1 Pi*MZ (rs28929474 C/T); HSD17B13 T/TA protector (Imputed DR2 1.00); CYP1A2 *1F A/C + rs2472297 C/C; ADORA2A rs5751876 C/C	Coffee is associated with less hepatic fibrosis, which weighs more with this genetic liver; most of the data is with caffeine coffee, but decaffeinated was also associated with lower liver enzymes. Decaffeinated avoids the effect of caffeine on sleep; filtered, without coffee (LDL).	Weak	chip + imputation	Chronic Inflammation; Metabolic Homeostasis decline; Sleep disturbance
Quality protein without high heme × predisposition to sarcopenia (support note; does not give ⊕ alone) 29 ⊕ foods	Fragility/sarcopenia score, ACTN3	Polygenic score of frailty at the 82nd percentile; sarcopenia "most likely"; ACTN3 R577X (rs1815739 T/C)	Genetic predisposition to lose lean mass: quality protein distributed by meals weighs more. The gene × protein interaction is not tested (it is risk stratification).	Weak	chip + polygenic score	Sarcopenia; Thin Mass Loss; Fragility

Also verified, without favourable indication: COMT Val158Val (rs4680 G/G; catechins and methylated flavanols faster) · UGT1A1 *1/*28 (rs887829 C/T; flavonoids in food dose without relevance) · CYP1A2 *1F A/C + rs2472297 C/C and ADORA2A C/C (caffeine) · AHR rs2066853 G/G · SLC22A4 L503F C/C (typical ergothioneine transport) · BCO1 R267S A/T without A379V (carotenoid conversion without practical relevance) · APOA2 −265 G/A, PPARG Pro12Pro, LIPC −514 C/C and APOE ε3/ε3 (typical fat response) · CYP3A5 *1/*3 (only for grapefruit). Support tolerances: Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence). Normal G6PD (without c.202G>A, rs1050828, or c.563C>T, rs5030868): faves without genetic risk of hemolysis (strong evidence). AGXT without G170R: no genetic predisposition to hyperoxaluria. ABCG2 Q141K absent (rs2231142 G/G): low predisposition to high uric acid.

3. Foods with favourable genetic indication (⊕), by category

In each category only ⊕ foods appear, ordered by the force of the indication (weighted sum of the mechanisms). The line of each category indicates how many foods were evaluated and how many were neutral.

3.1 Crucifers — 18 evaluated foods

Moderate ⊕: 16 · Weak ⊕: 2 · neutral (not listed): 0

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Bruxella Moderate ⊕	Of the most rich crucifers in glucosinolates; fiber and potassium. Oxidative stress; DNA Damage and DNA Repair Deficit; Chronic Inflammation; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); MTHFR, MTR, SLC19A1; PEMT, CHDH, PNPLA3, MBOAT7; AGT, ADD1, GNB3, CYP11B2; CETP, APOB, 9p21, ApoB polygenic Null GSTM1 prov. GSTA1 *B/*B; NRF2 -617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · low folate plus homocysteine with reduced MTHFR · choline compensates PEMT Met175 (liver) · potassium compensates the salt sensitivity · low LDL/ApoB (genetic profile with more to gain)	Several times a week; cut in half and cook little. Suggestion (not applied).	Moderate / Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype SNPs (4 chip + 1 imputed, r 0.92-0.998); chip; chip + imputation
Broccoli Moderate ⊕	Sulforaphane (active NRF2 and II phase), fiber, folate and vitamin K. Oxidative stress; Redox imbalance; DNA Damage and DNA Repair Deficit; Chronic Inflammation; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); MTHFR, MTR, SLC19A1; PEMT, CHDH, PNPLA3, MBOAT7; CFH, ARMS2 Null GSTM1 prov. GSTA1 *B/*B; NRF2 -617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T CFH rs1061170 C/T	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · low folate plus homocysteine with reduced MTHFR · choline compensates PEMT Met175 (liver) · lutein for macula (CFH 402H het.)	≥ 1 portion/day; steam 3-4 min or raw; add mustard in grain or raw rocket if cooked more. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype (4 chip + 1 imputed, r 0.92-0.998); chip
Cauliflower Moderate ⊕	Glucosinolates, fiber and choline with little energy. Oxidative stress; DNA Damage and DNA Repair Deficit; Chronic Inflammation; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); MTHFR, MTR, SLC19A1; PEMT, CHDH, PNPLA3, MBOAT7 Null GSTM1 prov. GSTA1 *B/*B; NRF2 -617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · low folate plus homocysteine with reduced MTHFR · choline compensates PEMT Met175 (liver)	Several times a week; short steam or roasted at moderate temperature. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype (4 chip + 1 imputed, r 0.92-0.998); chip

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Rocket Moderate ⓘ	Isothiocyanates + one of the largest sources of nitrate. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging; Arterial stiffness	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); NOS3; MTHFR, MTR, SLC19A1 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) NOS3 rs2070744 C/T; rs1799983 G/T MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · NO via nitrate (eNOS minor) · low folate plus homocysteine with reduced MTHFR	Daily in salads; raw on cooked crucifers. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype SNPs (4 chip + 1 imputed, r 0.92–0.998); chip + imputation; chip
Gulls Moderate ⓘ	Glucosinolates and folate. Oxidative stress; DNA Damage and DNA Repair Deficit; Epigenetic Changes; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); MTHFR, MTR, SLC19A1 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · low folate plus homocysteine with reduced MTHFR	Short jumps with olive oil and garlic. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype (4 chip + 1 imputed, r 0.92–0.998); chip
Crescent Moderate ⓘ	Phenylisothiocyanate (PEITC) and lutein. Oxidative stress; DNA Damage and DNA Repair Deficit; Chronic Inflammation; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); CFH, ARMS2 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) CFH rs1061170 C/T	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · lutein for macula (CFH 402H het.)	Raw in salad or soup together at the end. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype (4 chip + 1 imputed, r 0.92–0.998); chip
Cabbage Moderate ⓘ	Glucosinolates, lutein, calcium and vitamin K. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging; Osteoporosis	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); CFH, ARMS2 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) CFH rs1061170 C/T	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · lutein for macula (CFH 402H het.)	Several times a week (green broth with little salt or salt). Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype (4 chip + 1 imputed, r 0.92–0.998); chip
Kale Moderate ⓘ	Glucosinolates and the highest dose of lutein among kale. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); CFH, ARMS2 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) CFH rs1061170 C/T	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · lutein for macula (CFH 402H het.)	Raw massage or short jump. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype (4 chip + 1 imputed, r 0.92–0.998); chip
Black cabbage Moderate ⓘ	Glucosinolates + anthocyanins. Oxidative stress; Redox imbalance; DNA Damage and DNA Repair Deficit; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala	Raw in salad; preserves myrosinase. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype (4 chip + 1 imputed, r 0.92–0.998); chip

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Pak choi/Chinese kale Moderate ⓘ	Glucosinolates and nitrate. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); NOS3 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) NOS3 rs2070744 C/T; rs1799983 G/T	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · NO via nitrate (eNOS minor)	Short jump. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); haplotype of 5 SNPs concordant (4 chip + 1 imputed, <i>r</i> 0.92–0.998); chip + imputation
Radish Moderate ⓘ	Isothiocyanates and nitrate, raw. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21); NOS3 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) NOS3 rs2070744 C/T; rs1799983 G/T	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21 · NO via nitrate (eNOS minor)	Raw in salad, daily if you like. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); haplotype of 5 SNPs concordant (4 chip + 1 imputed, <i>r</i> 0.92–0.998); chip + imputation
Sweetheart cabbage Moderate ⓘ	Glucosinolates and fiber. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21) Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21	Raw in salad or cooked little. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype SNPs (4 chip + 1 imputed, <i>r</i> 0.92–0.998)
Savoy cabbage Moderate ⓘ	Glucosinolates and fiber. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21) Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21	To change with the other cabbages. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype SNPs (4 chip + 1 imputed, <i>r</i> 0.92–0.998)
Steaky cabbage Moderate ⓘ	Glucosinolates, fiber and vitamin C. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21) Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21	To change with the other cabbages. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype SNPs (4 chip + 1 imputed, <i>r</i> 0.92–0.998)
Turnip Moderate ⓘ	Glucosinolates with low energy. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21) Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21	In soups and roasts. Suggestion (not applied).	Moderate-weak chip + imputation (GSTM1 is CNV, inferred); concordant 5 haplotype SNPs (4 chip + 1 imputed, <i>r</i> 0.92–0.998)

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Cabbage Moderate ⊕	Glucosinolates and fiber; cheap and durable. Oxidative stress; DNA Damage and DNA Repair Deficit; Vascular Aging	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38; CDKN2B-AS1 (9p21) Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	isothiocyanates are longer; NRF2/ NQO1 intact to respond · Diet rich in vegetables/fruit attenuated the risk of 9p21	Raw in salad (red cabbage) or short salted. Suggestion (not applied).	Moderate-weak chip + imputation (<i>GSTM1 is CNV, inferred</i>); concordant 5 haplotype SNPs (4 chip + 1 imputed, <i>r 0.92–0.998</i>)
Broccoli sprouts Weak ⊕	The most concentrated source of glucoraphanin (10–100× broccoli). Oxidative stress; Redox imbalance; DNA Damage and DNA Repair Deficit; Chronic Inflammation	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1	isothiocyanates are longer; NRF2/ NQO1 intact to respond	A thin (20–30) raw, several times a week; chew well. Suggestion (not applied).	Moderate-weak chip + imputation (<i>GSTM1 is CNV, inferred</i>)
Raban / wasabi Weak ⊕	Very concentrated isothiocyanates (allyl-isothiocyanate). Oxidative stress; DNA Damage and DNA Repair Deficit; Chronic Inflammation	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1	isothiocyanates are longer; NRF2/ NQO1 intact to respond	Small amounts as condiment; helps to restore myrosinase in cooked crucifers. Suggestion (not applied).	Moderate-weak chip + imputation (<i>GSTM1 is CNV, inferred</i>)

3.2 Other vegetables, mushrooms and tubers — 30 evaluated foods

Moderate ⊕: 12 · Weak ⊕: 14 · neutral (not listed): 4

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Spinach Moderate ⊕	Folate, betaine, nitrate, lutein, magnesium and potassium. Epigenetic Changes; Vascular Aging; Arterial stiffness; Oxidative stress	CDKN2B-AS1 (9p21); NOS3; MTHFR, MTR, SLC19A1; BHMT, GNMT; CFH, ARMS2; AGT, ADD1, GNB3, CYP11B2; MTHFR 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) NOS3 rs2070744 C/T; rs1799983 G/T MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A BHMT rs3733890 G/A; GNMT rs9296404 T/T CFH rs1061170 C/T AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A MTHFR rs1801133 G/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · NO via nitrate (eNOS minor) · low folate plus homocysteine with reduced MTHFR · by BHMT of remethylation · lutein for macula (CFH 402H het.) · potassium compensates the salt sensitivity · B2 stabilizes the MTHFR 677T	Diary, raw or salted; fresh (histamine); the fiber of other foods compensates oxalate. Suggestion (not applied). <i>Observe in the intolerance report: Food Histamine.</i> <i>AGXT without G170R: no genetic predisposition to hyperoxaluria.</i>	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, <i>r 0.92–0.998</i>); chip + imputation; chip
Fresh peas Moderate ⊕	Fiber, folate, plant protein and lutein. Vascular Aging; Epigenetic Changes; Hyperglycemia and Intolerance to Glucose	CDKN2B-AS1 (9p21); MTHFR, MTR, SLC19A1; CFH, ARMS2; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CFH rs1061170 C/T CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C	Diet rich in vegetables/fruit attenuated the risk of 9p21 · low folate plus homocysteine with reduced MTHFR · lutein for macula (CFH 402H het.) · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/ MTNR1B)	Several times a week. Suggestion (not applied).	Moderate / Moderate-weak haplotype of 5 SNPs concordant (4 chip + 1 imputed, <i>r 0.92–0.998</i>); chip; chip + imputation

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Artichoke Moderate ⓘ	Inulin, fiber and potassium. Dysbiosis; Intestinal Barrier dysfunction; Vascular Aging	CDKN2B-AS1 (9p21); FUT2; MTHFR, MTR, SLC19A1; AGT, ADD1, GNB3, CYP11B2; CETP, APOB, 9p21, ApoB polygenic 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) FUT2 rs601338 G/A (secretor) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	Diet rich in vegetables/fruit attenuated the risk of 9p21 · secretor: more bifidobacteria with prebiotics · low folate plus homocysteine with reduced MTHFR · potassium compensates the salt sensitivity · low LDL/ApoB (genetic profile with more to gain)	Cooked or roasted; also in canned without salt. Suggestion (not applied).	Moderate / Moderate-weak <i>haplotype of 5 SNPs concordant (4 chip + 1 imputed, r 0.92–0.998); chip; chip + imputation</i>
Beet Moderate ⓘ	Nitrate, betaine (betalains) and folate. Vascular Aging; Arterial stiffness; Epigenetic Changes	CDKN2B-AS1 (9p21); NOS3; MTHFR, MTR, SLC19A1; BHMT, GNMT; AGT, ADD1, GNB3, CYP11B2 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) NOS3 rs2070744 C/T; rs1799983 G/T MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A BHMT rs3733890 G/A; GNMT rs9296404 T/T AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · NO via nitrate (eNOS minor) · low folate plus homocysteine with reduced MTHFR · by BHMT of remethylation · potassium compensates the salt sensitivity	Whole (cooked, roasted or raw grated); not using concentrated beet juices. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip + imputation; chip</i>
Okra Moderate ⓘ	Viscous mucilage (anti-LDL portfolio). Vascular Aging; Dysbiosis	CDKN2B-AS1 (9p21); CETP, APOB, 9p21, ApoB polygenic; MTHFR, MTR, SLC19A1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · low LDL/ApoB (genetic profile with more to gain) · low folate plus homocysteine with reduced MTHFR	Green or roasted. Suggestion (not applied).	Moderate / Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip + imputation; chip</i>
Algae Moderate ⓘ	Betaine, nitrate, potassium and magnesium. Epigenetic Changes; Vascular Aging; Arterial stiffness	CDKN2B-AS1 (9p21); NOS3; BHMT, GNMT; AGT, ADD1, GNB3, CYP11B2 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) NOS3 rs2070744 C/T; rs1799983 G/T BHMT rs3733890 G/A; GNMT rs9296404 T/T AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · NO via nitrate (eNOS minor) · by BHMT of remethylation · potassium compensates the salt sensitivity	Several times a week. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip + imputation; chip</i>
Lettuce Moderate ⓘ	Nitrate and folate (mainly Roman and dark leaves). Vascular Aging; Arterial stiffness; Epigenetic Changes	CDKN2B-AS1 (9p21); NOS3; MTHFR, MTR, SLC19A1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) NOS3 rs2070744 C/T; rs1799983 G/T MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · NO via nitrate (eNOS minor) · low folate plus homocysteine with reduced MTHFR	Daily basis of salad. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip + imputation; chip</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
French garlic Moderate ⓘ	Inulin and folate. Dysbiosis; Intestinal Barrier dysfunction; Epigenetic Changes	CDKN2B-AS1 (9p21); FUT2; MTHFR, MTR, SLC19A1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) FUT2 rs601338 G/A (secretor) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · secretor: more bifidobacteria with prebiotics · low folate plus homocysteine with reduced MTHFR	In soups and ovens. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Sweet potato Moderate ⓘ	Carotenoids, potassium and fiber. Oxidative stress; Vascular Aging; Hyperglycemia and Intolerance to Glucose	CDKN2B-AS1 (9p21); AGT, ADD1, GNB3, CYP11B2; TCF7L2, MTNR1B, GCKR 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C	Diet rich in vegetables/fruit attenuated the risk of 9p21 · potassium compensates the salt sensitivity · lower glycemic load (TCF7L2/MTNR1B)	Cooking (glucose index lower than roasted), preferably cooled and reheated. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Sparses Moderate ⓘ	Inulin and folate. Dysbiosis; Epigenetic Alterations	CDKN2B-AS1 (9p21); FUT2; MTHFR, MTR, SLC19A1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) FUT2 rs601338 G/A (secretor) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · secretor: more bifidobacteria with prebiotics · low folate plus homocysteine with reduced MTHFR	Several times a week at the time. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Aubergine Moderate ⓘ	Anti-LDL viscous fiber. Vascular Aging; Arterial stiffness	CDKN2B-AS1 (9p21); CETP, APOB, 9p21, ApoB polygenic 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	Diet rich in vegetables/fruit attenuated the risk of 9p21 · low LDL/ApoB (genetic profile with more to gain)	Roast or cooked with olive oil; fresh (histamine). Suggestion (not applied). <i>Observe in the intolerance report: Food Histamine.</i>	Moderate / Moderate-weak <i>haplotype of 5 SNPs concordant (4 chip + 1 imputed, r 0.92–0.998); chip + imputation</i>
Green bean Moderate ⓘ	Fiber and folate. Vascular Aging; Epigenetic Changes	CDKN2B-AS1 (9p21); MTHFR, MTR, SLC19A1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · low folate plus homocysteine with reduced MTHFR	Diary as follow-up. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Pumpkin Weak ⓘ	Carotenoids, fiber and potassium. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21); AGT, ADD1, GNB3, CYP11B2 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · potassium compensates the salt sensitivity	Roasted or in soup. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Apricot Weak ⓘ	Nitrate and apigenin in food dose. Vascular Aging; Arterial stiffness	CDKN2B-AS1 (9p21); NOS3 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) NOS3 rs2070744 C/T; rs1799983 G/T	Diet rich in vegetables/fruit attenuated the risk of 9p21 · NO via nitrate (eNOS minor)	Raw or in soups. Suggestion (not applied).	Moderate-weak <i>haplotype of 5 SNPs concordant (4 chip + 1 imputed, r 0.92–0.998); chip + imputation</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Garlic Weak ⊕	Organosulfurized compounds and prebiotic fiber. Dysbiosis; Vascular Aging; Chronic Inflammation	CDKN2B-AS1 (9p21); FUT2 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) FUT2 rs601338 G/A (secretor)	Diet rich in vegetables/fruit attenuated the risk of 9p21 · secretor: more bifidobacteria with prebiotics	Normal culinary use (not in supplement). Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Onion Weak ⊕	Prebiotic fiber (fructans) and quercetin in food dose. Dysbiosis; Intestinal Barrier dysfunction; Vascular Aging	CDKN2B-AS1 (9p21); FUT2 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) FUT2 rs601338 G/A (secretor)	Diet rich in vegetables/fruit attenuated the risk of 9p21 · secretor: more bifidobacteria with prebiotics	Daily, raw or cooked. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Endiveness / chicory Weak ⊕	Inulin and bitter compounds. Dysbiosis; Intestinal Barrier dysfunction	CDKN2B-AS1 (9p21); FUT2 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) FUT2 rs601338 G/A (secretor)	Diet rich in vegetables/fruit attenuated the risk of 9p21 · secretor: more bifidobacteria with prebiotics	Raw in salads. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Funk Weak ⊕	Nitrate and fiber. Vascular Aging; Arterial stiffness	CDKN2B-AS1 (9p21); NOS3 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) NOS3 rs2070744 C/T; rs1799983 G/T	Diet rich in vegetables/fruit attenuated the risk of 9p21 · NO via nitrate (eNOS minor)	Raw rolled or roasted. Suggestion (not applied).	Moderate-weak <i>haplotype of 5 SNPs concordant (4 chip + 1 imputed, r 0.92–0.998); chip + imputation</i>
Red pepper Weak ⊕	Vitamin C and carotenoids. Oxidative stress; Redox imbalance; Vascular Aging	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala	Raw in salads. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Paris mushroom Weak ⊕	Riboflavin, ergothioneine and potassium. Oxidative stress; Mitochondrial dysfunction	MTHFR; AGT, ADD1, GNB3, CYP11B2 MTHFR rs1801133 G/A AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A	B2 stabilizes the MTHFR 677T · potassium compensates the salt sensitivity	Several times a week. Suggestion (not applied).	Weak <i>chip</i>
Shiitake mushroom Weak ⊕	Riboflavin, ergothioneine, β-glucans and choline. Oxidative stress; Mitochondrial dysfunction; Immunosenescence	MTHFR; PEMT, CHDH, PNPLA3, MBOAT7 MTHFR rs1801133 G/A PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T	B2 stabilizes the MTHFR 677T · choline compensates PEMT Met175 (liver)	Several times a week; well cooked. Suggestion (not applied).	Weak <i>chip</i>
Carrot Weak ⊕	Carotenoids (normal conversion in practice) and fiber. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Raw or cooked; carotenoids instead of retinol. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)</i>
Courgette Weak ⊕	Fiber and water with low energy. Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Soup base and salted. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)</i>
Cucumber Weak ⊕	Hydration and fiber with low energy. Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Raw in salads. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Tomato Weak ⊕	Lycopene (more bioavailable cooked with olive oil) and potassium. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Cooked with extra virgin olive oil; already has genetically high lycopene, so normal food dose. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)</i>
Pleurotus mushroom Weak ⊕	Riboflavin, ergothioneine and β-glucans. Oxidative stress; Mitochondrial dysfunction; Immunosenescence	MTHFR MTHFR rs1801133 G/A	B2 stabilizes the MTHFR 677T	Several times a week. Suggestion (not applied).	Weak <i>chip</i>

3.3 Legumes and soybeans — 18 evaluated foods

Moderate ⊕: 18 · Weak ⊕: 0 · neutral (not listed): 0

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Beak grain Moderate ⊕	Fiber, folate, choline and plant protein. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis; Epigenetic Changes; Sarcopenia	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; AGT, ADD1, GNB3, CYP11B2; SLC30A8; PEMT, CHDH, PNPLA3, MBOAT7; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A SLC30A8 rs13266634 C/C; rs11558471 A/A PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics · potassium compensates the salt sensitivity · zinc associated with lower glucose with SLC30A8 A/A · choline compensates PEMT Met175 (liver) · protein against sarcopenia predisposition	1–2 portions/day; demolish well. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Lentils Moderate ⊕	Fiber, folate, plant protein, potassium and zinc. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis; Epigenetic Changes; Sarcopenia	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; AGT, ADD1, GNB3, CYP11B2; SLC30A8; PEMT, CHDH, PNPLA3, MBOAT7; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A SLC30A8 rs13266634 C/C; rs11558471 A/A PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics · potassium compensates the salt sensitivity · zinc associated with lower glucose with SLC30A8 A/A · choline compensates PEMT Met175 (liver) · protein against sarcopenia predisposition	1–2 portions/day; demolish or use germinated. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Edamame Moderate ⓘ	Complete protein, fiber, folate and choline. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Sarcopenia; Dysfunction of adipose tissue	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; PEMT, CHDH, PNPLA3, MBOAT7; AGT, ADD1, GNB3, CYP11B2; FADS1/ FADS2; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A FADS1 rs174546 C/C; FADS2 rs1535 A/A p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/ MTNR1B) · choline compensates PEMT Met175 (liver) · potassium compensates the salt sensitivity · efficient conversion of ALA to EPA · protein against sarcopenia predisposition	As input or in salads. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Soybean Moderate ⓘ	Complete protein, fiber, choline, isoflavones and ALA. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Sarcopenia; Dysfunction of adipose tissue	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; PEMT, CHDH, PNPLA3, MBOAT7; AGT, ADD1, GNB3, CYP11B2; FADS1/ FADS2; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A FADS1 rs174546 C/C; FADS2 rs1535 A/A p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/ MTNR1B) · choline compensates PEMT Met175 (liver) · potassium compensates the salt sensitivity · efficient conversion of ALA to EPA · protein against sarcopenia predisposition	Sometimes a week. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Dry peas Moderate ⓘ	Fiber and plant protein. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; AGT, ADD1, GNB3, CYP11B2; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/ MTNR1B) · secretor: more bifidobacteria with prebiotics · potassium compensates the salt sensitivity · protein against sarcopenia predisposition	In soups and purees. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Beans Moderate ⓘ	Fiber, folate and protein; L-DOPA natural. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis; Epigenetic Changes	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; AGT, ADD1, GNB3, CYP11B2; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/ MTNR1B) · secretor: more bifidobacteria with prebiotics · potassium compensates the salt sensitivity · protein against sarcopenia predisposition	At the time; without genetic restriction (normal G6PD). Suggestion (not applied). <i>Normal G6PD (without c.202G>A, rs1050828, or c.563C>T, rs5030868): faves without genetic risk of hemolysis (strong evidence).</i>	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Azuki beans Moderate ⓘ	Fiber and folate. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis; Epigenetic Changes	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; AGT, ADD1, GNB3, CYP11B2 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/ MTNR1B) · secretor: more bifidobacteria with prebiotics · potassium compensates the salt sensitivity	To change with the other beans. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
White bean Moderate ⓘ	Fiber and potassium. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; AGT, ADD1, GNB3, CYP11B2; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/ MTNR1B) · secretor: more bifidobacteria with prebiotics · potassium compensates the salt sensitivity · protein against sarcopenia predisposition	1–2 portions/day. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Embedded beans Moderate ⓘ	Fiber, folate and potassium. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis; Epigenetic Changes	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; AGT, ADD1, GNB3, CYP11B2; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/ MTNR1B) · secretor: more bifidobacteria with prebiotics · potassium compensates the salt sensitivity · protein against sarcopenia predisposition	1–2 portions/day; always well cooked. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Black bean Moderate ⓘ	Fiber, anthocyanins and folate. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis; Epigenetic Changes	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; AGT, ADD1, GNB3, CYP11B2; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics · potassium compensates the salt sensitivity · protein against sarcopenia predisposition	1–2 portions/day. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Beans Moderate ⓘ	Fiber and folate very high. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis; Epigenetic Changes	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; AGT, ADD1, GNB3, CYP11B2; frailty/sarcopenia score, ACTN3 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A p82 fragility; ACTN3 T/C	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics · potassium compensates the salt sensitivity · protein against sarcopenia predisposition	1–2 portions/day. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Humus Moderate ⓘ	Beak grain + tahini (zinc) + olive oil. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis	CETP, APOB, 9p21, ApoB polygenic; MTHFR, MTR, SLC19A1; TCF7L2, MTNR1B, GCKR; FUT2 CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor)	low LDL/ApoB (genetic profile with more to gain) · low folate plus homocysteine with reduced MTHFR · lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics	Homeless with little salt; with raw vegetables. Suggestion (not applied).	Moderate / Moderate-weak <i>chip + imputation; chip</i>
Mungo beans / soybean shoots Moderate ⓘ	Fiber and folate; also germinated. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis	MTHFR, MTR, SLC19A1; TCF7L2, MTNR1B, GCKR; FUT2 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor)	low folate plus homocysteine with reduced MTHFR · lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics	To change with the other beans. Suggestion (not applied).	Moderate-weak <i>chip</i>
Tempeh Moderate ⓘ	Fermented soybean: protein, fiber and riboflavin. Vascular Aging; Sarcopenia; Dysbiosis	CETP, APOB, 9p21, ApoB polygenic; PEMT, CHDH, PNPLA3, MBOAT7; MTHFR; frailty/sarcopenia score, ACTN3 CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T MTHFR rs1801133 G/A p82 fragility; ACTN3 T/C	low LDL/ApoB (genetic profile with more to gain) · choline compensates PEMT Met175 (liver) · B2 stabilizes the MTHFR 677T · protein against sarcopenia predisposition	Fresh, well conserved (histamine). Suggestion (not applied). <i>Observe in the intolerance report: Food Histamine.</i>	Moderate / Weak <i>chip + imputation; chip</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Peanut Moderate ⊕	Folate, monounsaturated fatty acids and arginine. Vascular Aging	MTHFR, MTR, SLC19A1; CETP, APOB, 9p21, ApoB polygenic MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	low folate plus homocysteine with reduced MTHFR · low LDL/ApoB (genetic profile with more to gain)	Natural, without salt (the roasted with salt are in another class). Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Lupins Moderate ⊕	Protein and fiber very high. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Sarcopenia	CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; frailty/sarcopenia score, ACTN3 CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C p82 fragility; ACTN3 T/C	low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/MTNR1B) · protein against sarcopenia predisposition	Demolize and pass through water to remove salt from the brine. Suggestion (not applied). <i>Observe in the intolerance report: Salt (genetic sensitivity to salt).</i>	Moderate / Moderate-weak <i>chip + imputation; chip</i>
Natto Moderate ⊕	Soybean fermented with vitamin K2. Vascular Aging; Osteoporosis; Sarcopenia	CETP, APOB, 9p21, ApoB polygenic; PEMT, CHDH, PNPLA3, MBOAT7; frailty/sarcopenia score, ACTN3 CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T p82 fragility; ACTN3 T/C	low LDL/ApoB (genetic profile with more to gain) · choline compensates PEMT Met175 (liver) · protein against sarcopenia predisposition	Small portions; fresh (histamine). Suggestion (not applied). <i>Observe in the intolerance report: Food Histamine.</i>	Moderate / Weak <i>chip + imputation; chip</i>
Tofu Moderate ⊕	Soybean protein (anti-LDL portfolio) and choline. Vascular Aging; Sarcopenia; Dysfunction of adipose tissue	CETP, APOB, 9p21, ApoB polygenic; PEMT, CHDH, PNPLA3, MBOAT7; frailty/sarcopenia score, ACTN3 CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T p82 fragility; ACTN3 T/C	low LDL/ApoB (genetic profile with more to gain) · choline compensates PEMT Met175 (liver) · protein against sarcopenia predisposition	Meat substitute several times a week. Suggestion (not applied).	Moderate / Weak <i>chip + imputation; chip</i>

3.4 Whole grain and pseudocereals — 18 evaluated foods

Moderate ⊕: 7 · Weak ⊕: 7 · neutral (not listed): 4

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Oat in flakes Moderate ⊕	β-glucan (low to LDL), prebiotic fiber and zinc. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis; Nutrient Sensing Deregulation	CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; SLC30A8 CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) SLC30A8 rs13266634 C/C; rs11558471 A/A	low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics · zinc associated with lower glucose with SLC30A8 A/A	Daily (40–60 g), without added sugar. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Moderate / Moderate-weak <i>chip + imputation; chip</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Oat meal Moderate ⓘ	The most concentrated form of β-glucan. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis	CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2; SLC30A8 CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) SLC30A8 rs13266634 C/C; rs11558471 A/A	low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics · zinc associated with lower glucose with SLC30A8 A/A	10–30 g/day in papaya or yogurt. Suggestion (not applied).	Moderate / Moderate-weak <i>chip + imputation; chip</i>
Wheat germ Moderate ⓘ	Betaine, folate, zinc, choline and spermidine. Epigenetic Changes; Autophagy / Deregulated Macroautophagy; Nutrient Sensing Deregulation	BHMT, GNMT; MTHFR, MTR, SLC19A1; SLC30A8; PEMT, CHDH, PNPLA3, MBOAT7; MTHFR BHMT rs3733890 G/A; GNMT rs9296404 T/T MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A SLC30A8 rs13266634 C/C; rs11558471 A/A PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T MTHFR rs1801133 G/A	by BHMT of remethylation · low folate plus homocysteine with reduced MTHFR · zinc associated with lower glucose with SLC30A8 A/A · choline compensates PEMT Met175 (liver) · B2 stabilizes the MTHFR 677T	1–2 tablespoons/day in yogurt or papaya. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Moderate-weak <i>chip + imputation; chip</i>
Barley / perlada Moderate ⓘ	β-glucan and very low glycemic index. Vascular Aging; Hyperglycemia and Intolerance to Glucose; Dysbiosis	CETP, APOB, 9p21, ApoB polygenic; TCF7L2, MTNR1B, GCKR; FUT2 CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor)	low LDL/ApoB (genetic profile with more to gain) · lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics	In soups, salads and instead of rice. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Moderate / Moderate-weak <i>chip + imputation; chip</i>
Quinoa Moderate ⓘ	Complete protein, betaine and choline. Epigenetic Changes; Hyperglycemia and Intolerance to Glucose; Sarcopenia	BHMT, GNMT; PEMT, CHDH, PNPLA3, MBOAT7; TCF7L2, MTNR1B, GCKR; MTHFR; frailty/ sarcopenia score, ACTN3 BHMT rs3733890 G/A; GNMT rs9296404 T/T PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C MTHFR rs1801133 G/A p82 fragility; ACTN3 T/C	by BHMT of remethylation · choline compensates PEMT Met175 (liver) · lower glycemic load (TCF7L2/MTNR1B) · B2 stabilizes the MTHFR 677T · protein against sarcopenia predisposition	Instead of white rice. Suggestion (not applied).	Moderate-weak <i>chip + imputation; chip</i>
Grain rye Moderate ⓘ	Arabinoxylans, lignans and low glycemic index. Hyperglycemia and Intolerance to Glucose; Dysbiosis; Epigenetic Changes	TCF7L2, MTNR1B, GCKR; FUT2; BHMT, GNMT TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) BHMT rs3733890 G/A; GNMT rs9296404 T/T	lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics · by BHMT of remethylation	In grains or flakes. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Moderate-weak <i>chip; chip + imputation</i>
Rye bread Moderate ⓘ	Lowest glycemic index bread; fiber. Hyperglycemia and Intolerance to Glucose; Dysbiosis	TCF7L2, MTNR1B, GCKR; FUT2; BHMT, GNMT TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C FUT2 rs601338 G/A (secretor) BHMT rs3733890 G/A; GNMT rs9296404 T/T	lower glycemic load (TCF7L2/MTNR1B) · secretor: more bifidobacteria with prebiotics · by BHMT of remethylation	Choosing 100% whole rye and with little salt. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Moderate-weak <i>chip; chip + imputation</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Wheat meal Weak ⊕	Insoluble fiber, betaine and zinc. Epigenetic Changes; Dysbiosis	BHMT, GNMT; SLC30A8; FUT2 BHMT rs3733890 G/A; GNMT rs9296404 T/T SLC30A8 rs13266634 C/C; rs11558471 A/A FUT2 rs601338 G/A (secretor)	by BHMT of remethylation · zinc associated with lower glucose with SLC30A8 A/A · secretor: more bifidobacteria with prebiotics	1–2 tablespoons/day. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Weak <i>chip + imputation; chip</i>
Amaranth Weak ⊕	Betaine and protein. Epigenetic Changes; Hyperglycemia and Intolerance to Glucose	BHMT, GNMT; TCF7L2, MTNR1B, GCKR BHMT rs3733890 G/A; GNMT rs9296404 T/T TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C	by BHMT of remethylation · lower glycemic load (TCF7L2/MTNR1B)	In grain or tufted without sugar. Suggestion (not applied).	Moderate-weak <i>chip + imputation; chip</i>
Bulgur Weak ⊕	Low glycemic index pre-cooked whole wheat. Hyperglycemia and Intolerance to Glucose; Epigenetic Changes	TCF7L2, MTNR1B, GCKR; BHMT, GNMT TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C BHMT rs3733890 G/A; GNMT rs9296404 T/T	lower glycemic load (TCF7L2/MTNR1B) · by BHMT of remethylation	In salads (table) and instead of rice. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Moderate-weak <i>chip; chip + imputation</i>
Espelta Weak ⊕	Old integral wheat. Hyperglycemia and Intolerance to Glucose; Epigenetic Changes	TCF7L2, MTNR1B, GCKR; BHMT, GNMT TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C BHMT rs3733890 G/A; GNMT rs9296404 T/T	lower glycemic load (TCF7L2/MTNR1B) · by BHMT of remethylation	In grain or as whole bread. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Moderate-weak <i>chip; chip + imputation</i>
Whole mass Weak ⊕	Fiber; low glycemic index al tooth. Hyperglycemia and Intolerance to Glucose	TCF7L2, MTNR1B, GCKR TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C	lower glycemic load (TCF7L2/MTNR1B)	Al dente; cooled and reheated gives more resistant starch. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Moderate-weak <i>chip</i>
Sarracene Weak ⊕	Rutin, fiber and low glycemic index. Hyperglycemia and Intolerance to Glucose; Vascular Aging	TCF7L2, MTNR1B, GCKR TCF7L2 rs7903146 C/T; MTNR1B rs10830963 C/G; GCKR T/C	lower glycemic load (TCF7L2/MTNR1B)	In grains or crepes without sugar. Suggestion (not applied).	Moderate-weak <i>chip</i>
Whole wheat bread Weak ⊕	Cereal fiber and betaine. Hyperglycemia and Intolerance to Glucose; Dysbiosis; Epigenetic Changes	BHMT, GNMT BHMT rs3733890 G/A; GNMT rs9296404 T/T	by BHMT of remethylation	Choosing whole 100% with little salt. Suggestion (not applied). <i>Without HLA-DQ2.5/DQ8 (rs2187668 C/C, rs7454108 T/T): very unlikely celiac disease; wheat, rye and barley without genetic restriction (strong evidence).</i>	Weak <i>chip + imputation</i>

3.5 Dry fruits and seeds — 16 evaluated foods

Moderate ⊕: 8 · Weak ⊕: 7 · neutral (not listed): 1

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Noz Moderate ⓘ	The nut with more ALA; polyphenols; low to LDL. Vascular Aging; Chronic Inflammation; Neuroinflammation	FADS1/FADS2; TCF7L2; CETP, APOB, 9p21, ApoB polygenic; MTHFR, MTR, SLC19A1 FADS1 rs174546 C/C; FADS2 rs1535 A/A TCF7L2 rs7903146 C/T, rs12255372 G/T CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	efficient conversion of ALA to EPA · Mediterranean diet attenuated the TCF7L2 effect · low LDL/ApoB (genetic profile with more to gain) · low folate plus homocysteine with reduced MTHFR	30 g/day (about 7 halves), without salt. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Linseeds Moderate ⓘ	ALA, woodwood and mucilage (viscous fiber). Vascular Aging; Chronic Inflammation; Dysbiosis	FADS1/FADS2; CETP, APOB, 9p21, ApoB polygenic; FUT2; MTHFR, MTR, SLC19A1 FADS1 rs174546 C/C; FADS2 rs1535 A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. FUT2 rs601338 G/A (secretor) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	efficient conversion of ALA to EPA · low LDL/ApoB (genetic profile with more to gain) · secretor: more bifidobacteria with prebiotics · low folate plus homocysteine with reduced MTHFR	1–2 tablespoons/day, ground at the time. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Cotton Moderate ⓘ	Monounsaturated, vitamin E and folate. Vascular Aging; Oxidative stress	TCF7L2; CETP, APOB, 9p21, ApoB polygenic; MTHFR, MTR, SLC19A1 TCF7L2 rs7903146 C/T, rs12255372 G/T CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	Mediterranean diet attenuated the TCF7L2 effect · low LDL/ApoB (genetic profile with more to gain) · low folate plus homocysteine with reduced MTHFR	30 g/day, natural. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Almond Moderate ⓘ	Monounsaturated, vitamin E, riboflavin and fiber. Vascular Aging; Oxidative stress	TCF7L2; CETP, APOB, 9p21, ApoB polygenic; MTHFR TCF7L2 rs7903146 C/T, rs12255372 G/T CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. MTHFR rs1801133 G/A	Mediterranean diet attenuated the TCF7L2 effect · low LDL/ApoB (genetic profile with more to gain) · B2 stabilizes the MTHFR 677T	30 g/day, natural. Suggestion (not applied).	Moderate / Moderate-weak <i>chip; chip + imputation</i>
Pistachio Moderate ⓘ	Lutein, potassium and fiber. Vascular Aging; Oxidative stress	CETP, APOB, 9p21, ApoB polygenic; AGT, ADD1, GNB3, CYP11B2; CFH, ARMS2 CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom. AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A CFH rs1061170 C/T	low LDL/ApoB (genetic profile with more to gain) · potassium compensates the salt sensitivity · lutein for macula (CFH 402H het.)	30 g/day, without salt. Suggestion (not applied).	Moderate / Weak <i>chip + imputation; chip</i>
Hemp seeds Moderate ⓘ	ALA, zinc and complete protein. Vascular Aging; Chronic Inflammation; Insulin resistance	FADS1/FADS2; SLC30A8; MTHFR, MTR, SLC19A1; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C; FADS2 rs1535 A/A SLC30A8 rs13266634 C/C; rs11558471 A/A MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A p82 fragility; ACTN3 T/C	efficient conversion of ALA to EPA · zinc associated with lower glucose with SLC30A8 A/A · low folate plus homocysteine with reduced MTHFR · protein against sarcopenia predisposition	1–2 tablespoons/day. Suggestion (not applied).	Moderate-weak <i>chip</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Chia seeds Moderate ⊕	ALA and viscous fiber. Vascular Aging; Chronic Inflammation; Dysbiosis	FADS1/FADS2; CETP, APOB, 9p21, ApoB polygenic FADS1 rs174546 C/C; FADS2 rs1535 A/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	efficient conversion of ALA to EPA · low LDL/ApoB (genetic profile with more to gain)	1–2 tablespoons/day, hydrated. Suggestion (not applied).	Moderate / Moderate- weak <i>chip; chip + imputation</i>
Cashew Moderate ⊕	Zinc and magnesium. Insulin resistance; Immunosenescence	SLC30A8; CETP, APOB, 9p21, ApoB polygenic SLC30A8 rs13266634 C/C; rs11558471 A/A CETP rs708272 G/G + rs3764261 C/ C; APOB rs693 G/A; 9p21 hom.	zinc associated with lower glucose with SLC30A8 A/A · low LDL/ApoB (genetic profile with more to gain)	To change, without salt. Suggestion (not applied).	Moderate / Weak <i>chip; chip + imputation</i>
Macadamia nut Weak ⊕	Monounsaturated (such as olive oil). Vascular Aging	CETP, APOB, 9p21, ApoB polygenic CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	low LDL/ApoB (genetic profile with more to gain)	To change, natural. Suggestion (not applied).	Moderate <i>chip + imputation</i>
Pecan Weak ⊕	Polyphenols and monounsaturated. Vascular Aging	CETP, APOB, 9p21, ApoB polygenic CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	low LDL/ApoB (genetic profile with more to gain)	To change, natural. Suggestion (not applied).	Moderate <i>chip + imputation</i>
Pumpkin seeds Weak ⊕	The best vegetable sources of zinc and magnesium. Insulin resistance; Immunosenescence; Thymic Involution	SLC30A8; AGT, ADD1, GNB3, CYP11B2 SLC30A8 rs13266634 C/C; rs11558471 A/A AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A	zinc associated with lower glucose with SLC30A8 A/A · potassium compensates the salt sensitivity	30 g/day, without salt. Suggestion (not applied).	Weak <i>chip</i>
Sunflower seeds Weak ⊕	Folate and vitamin E (very linoleic acid). Epigenetic Changes; Oxidative stress	MTHFR, MTR, SLC19A1 MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	low folate plus homocysteine with reduced MTHFR	Small portions, alternating with ALA-rich seeds. Suggestion (not applied).	Moderate-weak <i>chip</i>
Pine nut Weak ⊕	Zinc and monounsaturated. Insulin resistance; Immunosenescence	SLC30A8 SLC30A8 rs13266634 C/C; rs11558471 A/A	zinc associated with lower glucose with SLC30A8 A/A	Small portions. Suggestion (not applied).	Weak <i>chip</i>
Sesame seeds Weak ⊕	Zinc, calcium and firewood. Insulin resistance; Immunosenescence; Osteoporosis	SLC30A8 SLC30A8 rs13266634 C/C; rs11558471 A/A	zinc associated with lower glucose with SLC30A8 A/A	Sprinkler; also as tahini. Suggestion (not applied).	Weak <i>chip</i>
Tahini Weak ⊕	Zinc and calcium from sesame. Insulin resistance; Immunosenescence; Osteoporosis	SLC30A8 SLC30A8 rs13266634 C/C; rs11558471 A/A	zinc associated with lower glucose with SLC30A8 A/A	1 Spoon in sauces and humus. Suggestion (not applied).	Weak <i>chip</i>

3.6 Fish and shellfish — 16 evaluated foods

Moderate ⊕: 3 · Weak ⊕: 9 · neutral (not listed): 4

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Horse Moderate ⓘ	Small fatty fish, with EPA/DHA and riboflavin. Inflammaging; Chronic Inflammation; Vascular Aging; Neuroinflammation	FADS1/FADS2, ELOVL2, PNPLA3, IL6; PEMT, CHDH, PNPLA3, MBOAT7; MTHFR; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T MTHFR rs1801133 G/A p82 fragility; ACTN3 T/C	EPA/DHA preformed equilibrium high FADS · choline compensates PEMT Met175 (liver) · B2 stabilizes the MTHFR 677T · protein against sarcopenia predisposition	2-3×/week alternating; very fresh (histamine). Suggestion (not applied). <i>Observe in the intolerance report: Food Histamine.</i>	Moderate-weak <i>chip</i>
Fresh salmon Moderate ⓘ	EPA/DHA, astaxanthin and vitamin D. Inflammaging; Chronic Inflammation; Vascular Aging; Neuroinflammation	FADS1/FADS2, ELOVL2, PNPLA3, IL6; PEMT, CHDH, PNPLA3, MBOAT7; AGT, ADD1, GNB3, CYP11B2; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A p82 fragility; ACTN3 T/C	EPA/DHA preformed equilibrium high FADS · choline compensates PEMT Met175 (liver) · potassium compensates the salt sensitivity · protein against sarcopenia predisposition	1-2×/week; in the oven or steam. Suggestion (not applied).	Moderate-weak <i>chip</i>
Fresh sardine Moderate ⓘ	EPA/DHA, vitamin D, choline and calcium (spinhas). Inflammaging; Chronic Inflammation; Vascular Aging; Neuroinflammation; Sarcopenia	FADS1/FADS2, ELOVL2, PNPLA3, IL6; PEMT, CHDH, PNPLA3, MBOAT7; AGT, ADD1, GNB3, CYP11B2; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A p82 fragility; ACTN3 T/C	EPA/DHA preformed equilibrium high FADS · choline compensates PEMT Met175 (liver) · potassium compensates the salt sensitivity · protein against sarcopenia predisposition	2-3×/week; in the oven, cooked or grilled without carbonization. Suggestion (not applied). <i>ABCG2 Q141K absent (rs2231142 G/G): low predisposition to high uric acid.</i>	Moderate-weak <i>chip</i>
Fresh herb Weak ⓘ	EPA/DHA and vitamin D. Inflammaging; Chronic Inflammation; Vascular Aging; Neuroinflammation	FADS1/FADS2, ELOVL2, PNPLA3, IL6; PEMT, CHDH, PNPLA3, MBOAT7; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T p82 fragility; ACTN3 T/C	EPA/DHA preformed equilibrium high FADS · choline compensates PEMT Met175 (liver) · protein against sarcopenia predisposition	To change; very fresh (histamine). Suggestion (not applied). <i>Observe in the intolerance report: Food Histamine.</i>	Moderate-weak <i>chip</i>
Carapao Weak ⓘ	Small fatty fish, with little mercury. Inflammaging; Chronic Inflammation; Vascular Aging; Neuroinflammation	FADS1/FADS2, ELOVL2, PNPLA3, IL6; PEMT, CHDH, PNPLA3, MBOAT7; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T p82 fragility; ACTN3 T/C	EPA/DHA preformed equilibrium high FADS · choline compensates PEMT Met175 (liver) · protein against sarcopenia predisposition	To change with sardine and horse mackerel. Suggestion (not applied).	Moderate-weak <i>chip</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Oyster Weak ⊕	The largest zinc source; some EPA/DHA. Insulin resistance; Immunosenescence; Thymic Involution	SLC30A8; FADS1/FADS2, ELOVL2, PNPLA3, IL6 SLC30A8 rs13266634 C/C; rs11558471 A/A FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G	zinc associated with lower glucose with SLC30A8 A/A · EPA/DHA preformed equilibrium high FADS	Occasionally (1–2x/month): they have moderate iron. Suggestion (not applied).	Moderate-weak <i>chip</i>
Trout Weak ⊕	EPA/DHA with little mercury. Inflammaging; Chronic Inflammation; Vascular Aging	FADS1/FADS2, ELOVL2, PNPLA3, IL6; PEMT, CHDH, PNPLA3, MBOAT7; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T p82 fragility; ACTN3 T/C	EPA/DHA preformed equilibrium high FADS · choline compensates PEMT Met175 (liver) · protein against sarcopenia predisposition	To change with the other fatty fish. Suggestion (not applied).	Moderate-weak <i>chip</i>
Anchovy / fresh anchovy Weak ⊕	Small fatty fish. Inflammaging; Chronic Inflammation; Vascular Aging	FADS1/FADS2, ELOVL2, PNPLA3, IL6; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G p82 fragility; ACTN3 T/C	EPA/DHA preformed equilibrium high FADS · protein against sarcopenia predisposition	Fresh, fried in little oil or in homemade scale with little salt. Suggestion (not applied).	Moderate-weak <i>chip</i>
Golden Weak ⊕	Moderate Omega-3 (mainly aquaculture). Inflammaging; Vascular Aging; Sarcopenia	FADS1/FADS2, ELOVL2, PNPLA3, IL6; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G p82 fragility; ACTN3 T/C	EPA/DHA preformed equilibrium high FADS · protein against sarcopenia predisposition	In the oven or grilled without carbonization. Suggestion (not applied).	Moderate-weak <i>chip</i>
Sea bass Weak ⊕	Moderate Omega-3. Inflammaging; Vascular Aging; Sarcopenia	FADS1/FADS2, ELOVL2, PNPLA3, IL6; frailty/sarcopenia score, ACTN3 FADS1 rs174546 C/C, rs174537 G/G; FADS2 rs1535 A/A; PNPLA3 C/G p82 fragility; ACTN3 T/C	EPA/DHA preformed equilibrium high FADS · protein against sarcopenia predisposition	In the oven or grill without carbonization. Suggestion (not applied).	Moderate-weak <i>chip</i>
Fresh / frozen cod Weak ⊕	Thin protein and choline, without salt. Sarcopenia; Thin Mass Loss; Metabolic Homeostasis Decline	PEMT, CHDH, PNPLA3, MBOAT7; frailty/sarcopenia score, ACTN3 PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T p82 fragility; ACTN3 T/C	choline compensates PEMT Met175 (liver) · protein against sarcopenia predisposition	Fresh or frozen (the salty one enters the salt account). Suggestion (not applied).	Weak <i>chip</i>
Fish Weak ⊕	Thin protein and choline, without saturated fat. Sarcopenia; Thin Mass Loss; Metabolic Homeostasis Decline	PEMT, CHDH, PNPLA3, MBOAT7; frailty/sarcopenia score, ACTN3 PEMT rs7946 C/T; PNPLA3 C/G; MBOAT7 C/T p82 fragility; ACTN3 T/C	choline compensates PEMT Met175 (liver) · protein against sarcopenia predisposition	Base protein in several meals. Suggestion (not applied).	Weak <i>chip</i>

3.7 Red fruits and other fruit — 26 evaluated foods

Moderate ⊕: 9 · Weak ⊕: 17 · neutral (not listed): 0

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Avocado Moderate ⓘ	Monounsaturated, potassium, folate and fiber. Vascular Aging; Oxidative stress	CDKN2B-AS1 (9p21); AGT, ADD1, GNB3, CYP11B2; MTHFR, MTR, SLC19A1; CFH, ARMS2; CETP, APOB, 9p21, ApoB polygenic 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A CFH rs1061170 C/T CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	Diet rich in vegetables/fruit attenuated the risk of 9p21 · potassium compensates the salt sensitivity · low folate plus homocysteine with reduced MTHFR · lutein for macula (CFH 402H het.) · low LDL/ApoB (genetic profile with more to gain)	1/2 per day; fresh (histamine). Suggestion (not applied). <i>Observe in the intolerance report: Food Histamine.</i>	Moderate / Moderate-weak <i>haplotype of 5 SNPs concordant (4 chip + 1 imputed, r 0.92–0.998); chip; chip + imputation</i>
Raspberries Moderate ⓘ	Elagitaninos and the largest fiber of berries. Oxidative stress; Vascular Aging; Dysbiosis	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP; CETP, APOB, 9p21, ApoB polygenic 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala · low LDL/ApoB (genetic profile with more to gain)	Daily, fresh or frozen. Suggestion (not applied).	Moderate / Moderate-weak <i>haplotype of 5 SNPs concordant (4 chip + 1 imputed, r 0.92–0.998); chip; chip + imputation</i>
Orange Moderate ⓘ	Vitamin C, pectin and flavanones. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP; CETP, APOB, 9p21, ApoB polygenic 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala · low LDL/ApoB (genetic profile with more to gain)	Whole (non-juice); vitamin C at meals can increase iron absorption. Suggestion (not applied).	Moderate / Moderate-weak <i>haplotype of 5 SNPs concordant (4 chip + 1 imputed, r 0.92–0.998); chip; chip + imputation</i>
Strawberries Moderate ⓘ	Vitamin C, folate and fisetin in food dose. Oxidative stress; Vascular Aging; Cell Senescence	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP; MTHFR, MTR, SLC19A1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala · low folate plus homocysteine with reduced MTHFR	Dairy at the time. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Banana Moderate ⓘ	Potassium; resistant starch when not mature. Vascular Aging; Dysbiosis	CDKN2B-AS1 (9p21); AGT, ADD1, GNB3, CYP11B2; FUT2 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A FUT2 rs601338 G/A (secretor)	Diet rich in vegetables/fruit attenuated the risk of 9p21 · potassium compensates the salt sensitivity · secretor: more bifidobacteria with prebiotics	Little mature (more resistant starch, less sugar). Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Kiwi Moderate ⓘ	Vitamin C, potassium and fiber. Oxidative stress; Vascular Aging; Sleep disturbance	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP; AGT, ADD1, GNB3, CYP11B2 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A AGT rs699 A/G; ADD1 rs4961 G/T; GNB3 rs5443 C/T; CYP11B2 A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala · potassium compensates the salt sensitivity	1–2/day.	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Apple Moderate ⊕	Pectin and polyphenols. Vascular Aging; Dysbiosis	CDKN2B-AS1 (9p21); CETP, APOB, 9p21, ApoB polygenic 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	Diet rich in vegetables/fruit attenuated the risk of 9p21 · low LDL/ApoB (genetic profile with more to gain)	Whole, with bark. Suggestion (not applied).	Moderate / Moderate-weak <i>haplotype of 5 SNPs concordant (4 chip + 1 imputed, r 0.92–0.998); chip + imputation</i>
Pera Moderate ⊕	Pectin and fiber. Vascular Aging; Dysbiosis	CDKN2B-AS1 (9p21); CETP, APOB, 9p21, ApoB polygenic 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) CETP rs708272 G/G + rs3764261 C/C; APOB rs693 G/A; 9p21 hom.	Diet rich in vegetables/fruit attenuated the risk of 9p21 · low LDL/ApoB (genetic profile with more to gain)	Whole, with bark. Suggestion (not applied).	Moderate / Moderate-weak <i>haplotype of 5 SNPs concordant (4 chip + 1 imputed, r 0.92–0.998); chip + imputation</i>
Papaya Moderate ⊕	Carotenoids and folate. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21); MTHFR, MTR, SLC19A1 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) MTHFR rs1801133 G/A + rs1801131 T/G; MTR A/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · low folate plus homocysteine with reduced MTHFR	Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Blackberries Weak ⊕	Anthocyanins and fiber. Oxidative stress; Vascular Aging; Dysbiosis	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala	Daily, fresh or frozen. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Blueberries (fresh) Weak ⊕	Proanthocyanidins. Oxidative stress; Dysbiosis	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala	Fresh or frozen, without sugar. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Guava Weak ⊕	The fruit with more vitamin C and fiber. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala	Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Currants Weak ⊕	Vitamin C and anthocyanins. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala	Fresh or frozen. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>
Blueberries Weak ⊕	Anthocyanins; low sugar and fiber. Oxidative stress; Redox imbalance; Vascular Aging; Neuroinflammation	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala	Daily (1 cup), fresh or frozen. Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip</i>

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Pomegranate Weak ⊕	Elagitannins (precursors of urolitins). Oxidative stress; Mitochondrial Quality Control; Vascular Aging	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala	Whole bagasses (non-juice). Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip
Tangerine / clementin Weak ⊕	Vitamin C and carotenoids. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21); SOD2, GPX1, SELENOP 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed) SOD2 rs4880 G/G; GPX1 rs1050450 G/A	Diet rich in vegetables/fruit attenuated the risk of 9p21 · Food antioxidants with SOD2 Ala/Ala	Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998); chip
Alperce / apricot Weak ⊕	Carotenoids. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Fresh, whole (dry is another class). Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)
Plum Weak ⊕	Polyphenols and fiber. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Fresh, whole. Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)
Cherries Weak ⊕	Anthocyanins and melatonin in food dose. Oxidative Stress; Chronic Inflammation	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	At the time, as a dessert. Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)
Diospirus Weak ⊕	Carotenoids and tannins. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Fresh, moderate portion. Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)
Fig Weak ⊕	Fiber and polyphenols. Oxidative stress; Dysbiosis	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Fresh, whole. Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)
Ginjas Weak ⊕	Anthocyanins. Oxidative Stress; Chronic Inflammation	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Fresh or frozen, without sugar. Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)
Mango Weak ⊕	Carotenoids. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Moderate portion (more sugar). Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)
Watermelon Weak ⊕	Lycopene and citrulline. Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Moderate portion. Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)
Peach Weak ⊕	Carotenoids and fiber. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Fresh, whole. Suggestion (not applied).	Moderate-weak 5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Grapes Weak ⊕	Resveratrol and polyphenols of the bark. Oxidative stress; Vascular Aging	CDKN2B-AS1 (9p21) 9p21: homozygous risk haplotype of 5 SNPs (4 chip + 1 imputed)	Diet rich in vegetables/fruit attenuated the risk of 9p21	Moderate portion (more sugar). Suggestion (not applied).	Moderate-weak <i>5 concordant haplotype (4 chip + 1 imputed, r 0.92–0.998)</i>

3.8 Olive oil and other fats — 3 evaluated foods

Moderate ⊕: 0 · Weak ⊕: 3 · neutral (not listed): 0

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Extra virgin olive Weak ⊕	Monounsaturated and polyphenols (hydroxytyrosol, oleocanthal). Vascular Aging; Oxidative stress; Inflammaging; Hyperglycemia and Intolerance to Glucose	TCF7L2; PON1 TCF7L2 rs7903146 C/T, rs12255372 G/T PON1 Q192Q (rs662 T/T), M55M	Mediterranean diet attenuated the TCF7L2 effect · polyphenols of olive oil up to low PON1	40–50 mL/day as the main fat, raw and for cooking at moderate temperature. Suggestion (not applied).	Moderate-weak <i>chip; chip + imputation</i>
Rape oil/canola Weak ⊕	ALA and monounsaturated. Inflammaging; Vascular Aging	FADS1/FADS2 FADS1 rs174546 C/C; FADS2 rs1535 A/A	efficient conversion of ALA to EPA	Alternative to cooking, but olive oil remains the main fat. Suggestion (not applied).	Moderate-weak <i>chip</i>
Linseed oil Weak ⊕	The oil with more ALA. Inflammaging; Vascular Aging	FADS1/FADS2 FADS1 rs174546 C/C; FADS2 rs1535 A/A	efficient conversion of ALA to EPA	1 raw soup (never warm), stored in the cold. Suggestion (not applied).	Moderate-weak <i>chip</i>

3.9 Fermented dairy products and other fermented foods — 7 evaluated

Moderate ⊕: 0 · Weak ⊕: 3 · neutral (not listed): 4

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Greek yogurt Weak ⊕	More protein per portion. Dysbiosis; Osteoporosis; Sarcopenia	LCT/MCM6, HFE; MTHFR; frailty/ sarcopenia score, ACTN3 LCT rs4988235 G/A; HFE rs1799945 C/G MTHFR rs1801133 G/A p82 fragility; ACTN3 T/C	digests lactose; calcium reduces iron absorption · B2 stabilizes the MTHFR 677T · protein against sarcopenia predisposition	Thin version, without sugar. Suggestion (not applied).	Weak <i>chip (2 SNPs); chip</i>
Natural yogurt Weak ⊕	Protein, calcium and living bacteria. Dysbiosis; Osteoporosis; Sarcopenia	LCT/MCM6, HFE; MTHFR; frailty/ sarcopenia score, ACTN3 LCT rs4988235 G/A; HFE rs1799945 C/G MTHFR rs1801133 G/A p82 fragility; ACTN3 T/C	digests lactose; calcium reduces iron absorption · B2 stabilizes the MTHFR 677T · protein against sarcopenia predisposition	1–2/day at meals, lean and without sugar. Suggestion (not applied).	Weak <i>chip (2 SNPs); chip</i>
Milk Kefir Weak ⊕	Large variety of living microorganisms. Dysbiosis; Intestinal Barrier dysfunction; Osteoporosis	LCT/MCM6, HFE; MTHFR LCT rs4988235 G/A; HFE rs1799945 C/G MTHFR rs1801133 G/A	digests lactose; calcium reduces iron absorption · B2 stabilizes the MTHFR 677T	1 cup at meals; lean (histamine). Suggestion (not applied). <i>Observe in the intolerance report: Food Histamine.</i>	Weak <i>chip (2 SNPs); chip</i>

3.10 Coffee, tea and cocoa — 8 evaluated foods

Moderate ⊕: 0 · Weak ⊕: 4 · neutral (not listed): 4

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Decaffeinated coffee Weak ⊕	Chlorogenic acids and coffee polyphenols without caffeine. Chronic Inflammation; Metabolic Homeostasis decline; Sleep disturbance	PNPLA3, MBOAT7, SERPINA1, HSD17B13, CYP1A2, ADORA2A PNPLA3 C/G; SERPINA1 C/T; CYP1A2 rs762551 A/C; ADORA2A C/C	coffee protects the liver; without caffeine (by sleep)	Filtered in paper; at ease until the afternoon. Suggestion (not applied).	Weak chip + imputation
White tea Weak ⊕	Catechins with less caffeine. Oxidative stress	HFE, TMPRSS6 HFE H63D rs1799945 C/G	tannins at meals reduce iron (H63D)	The morning meals. Suggestion (not applied).	Weak chip
Black tea Weak ⊕	Theaflavins. Oxidative stress; Vascular Aging	HFE, TMPRSS6 HFE H63D rs1799945 C/G	tannins at meals reduce iron (H63D)	At morning meals (caffeine count of the day). Suggestion (not applied). <i>Observe in the intolerance report: Coffee.</i>	Weak chip
Green tea Weak ⊕	Catechins and L-theanine. Oxidative stress; Vascular Aging	HFE, TMPRSS6 HFE H63D rs1799945 C/G	tannins at meals reduce iron (H63D)	At morning meals (caffeine count, ≤ 200 mg); without extracts. Suggestion (not applied). <i>Observe in the intolerance report: Coffee.</i>	Weak chip

3.11 Herbs, spices and condiments — 11 evaluated foods

Moderate ⊕: 0 · Weak ⊕: 2 · neutral (not listed): 9

Food	Why to longevity	Genes · genotype	Predicted genetic response	Suggestion (not applied)	Evidence · trust
Horseradish Weak ⊕	Very concentrated isothiocyanates. Oxidative stress; DNA Damage and DNA Repair Deficit	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1	isothiocyanates are longer; NRF2/ NQO1 intact to respond	Small amounts as condiment. Suggestion (not applied).	Moderate-weak chip + imputation (<i>GSTM1 is CNV, inferred</i>)
Mustard seeds Weak ⊕	Active myrosinase: restores sulforaphane in cooked crucifers. Oxidative stress; DNA Damage and DNA Repair Deficit	GSTM1, GSTA1, NFE2L2, NQO1, GSTP1, TAS2R38 Null GSTM1 prov. GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1	isothiocyanates are longer; NRF2/ NQO1 intact to respond	1 mustard teaspoon in ground grain on cooked crucifers. Suggestion (not applied).	Moderate-weak chip + imputation (<i>GSTM1 is CNV, inferred</i>)

3.12 Eggs — 2 evaluated foods

Moderate ⊕: 0 · Weak ⊕: 0 · neutral (not listed): 2

Abstract: 141 of the evaluated longevity 173 foods have favourable genetic indication (moderate 73 ⊕, weak 68 ⊕); 32 are neutral and not listed.

4. Limitations

- **The nutritional evidence is, in general, weak to moderate.** The interactions used come from case-control studies, cohorts and secondary analyses of trials, with small effects and incomplete replication (e.g., the interaction 9p21 × prudent diet comes from two cohorts in the same study; that of TCF7L2 × Mediterranean diet, from a secondary analysis of an trial; that of GSTM1 × crucifers has contradictory results). None is a formal clinical recommendation.
- **"⊕" is not "obligatory" or "neutral" is "bad".** Neutral foods remain useful for longevity; there is only one genetic variant that increases the benefit. The effect of the global dietary pattern is much greater than that of any variant.
- **Fiber, LDL/ApoB and potassium** have strong benefit in everyone; the genetics here only indicates that there is more to gain, not that the answer is different.
- **Chip + imputation, no sequencing.** Only the variants measured or imputed with DR2 $\geq$ 0.8 are seen. GSTM1 (deletion) is inferred; GSTT1 and AMY1 are not testable; there is no microbiome data, which greatly changes the response to polyphenols and fiber.
- **Consistency with the other reports:** Foods in "limit" in the intolerance report were removed; those in "watching" have the note on the line. Iron: ferritin target below 100, so very rich in heme iron does not enter. The fish oil that already takes does not change: the fatty fish complements it.
- **Doses of supplements are not the subject of this report:** The same compounds in extract (sulforaphane, curcumin, catechins, quercetin) have another risk profile and follow the supplement report.
- **It is not a diagnosis.** Only genetics: no analyses or exams were used. All indications are suggestion (not applied).

5. Sources and methods

- **Method:** The same as the intolerant report (list cured → mechanisms by food → chip genotypes and imputation → class). Each food receives the favourable mechanisms that apply to its composition; weighting by evidence (moderate 2, moderate-weak 1.5, weak 1). Moderate ⊕ = at least one moderate-weak mechanism and sum $\geq$ 3; Weak ⊕ : remaining with at least one mechanism. The support note (protein × sarcopenia) and the tolerances do not give ⊕ alone.
- **Genotypes:** Genotyping chip (GRCh38) as primary source; Beagle 5.5 imputation with a public reference panel of complete genomes to 30x (only DR2 $\geq$ 0.8); polygenic scores calibrated in the reference population.
- **Ageing mechanisms:** The list of 84 mechanisms of longevity objectives; each food cites the ones that touch the most.
- **Background:** Crucifers × GSTM1 (Lin 1998; Spitz 2000; Joseph 2004; Gasper 2005; Lampe 2009 review); sulforaphane/NRF2 (Rioud 2009; Egner 2014; 9p21 × prudent diet (Do 2011); TCF7L2 × Mediterranean diet (Corella 2013); fiber × TCF7L2 (Hindy 2016); diet portfolio and β -glucan (Jenkins 2003; Whitehead 2014; Ho 2016); PON1 × olive oil (Fernandez-Castillejo 2017); FADS and conversion of ALA (Gillingham 2013); PNPLA3 × omega-6/omega-3 (Santoro 2012); folate × MTHFR (Jacques 1996); riboflavin × MTHFR (McNulty 2006; Wilson 2012); PEMT and choline (from Costa 2006); betaine (Olthof 2003); potassium and salt sensitivity (GenSalt, Gu 2007; consensus AHA 2016); food nitrate (Webb 2008; Kapil 2015); zinc × SLC30A8 (Kanoni 2011); lutein / zeaxanthin (AREDS2 2013); SOD2 × fruit and vegetables (Ambrosone et al.); tea and iron absorption (Hurrell et al.); coffee and liver (Wijanpreecha 2016; Xiao 2014, decaf; COMT × green tea (Wu 2003).